Cross-Protocol Stablecoin Stress Propagation: A Provenance-Aware AMM Network Analysis of Curve Finance and Uniswap v3

Anonymous Author(s)

ABSTRACT

Stablecoin stress episodes generate observable liquidity flows across automated market maker (AMM) protocols, yet existing empirical analyses rarely distinguish between within-protocol co-movement and cross-protocol propagation, or between execution-grade on-chain data and public market context. This paper addresses both gaps. We introduce a *provenance-aware claim gate* that assigns Tier-A (execution-grade on-chain) or Tier-B (public market context) status to every node and feature, then filters each empirical edge by data tier and statistical significance before permitting a paper-level claim.

Applying this framework to the June 2023 USDT/Curve event across Curve Finance and Uniswap v3—both Tier-A AMM protocols—we find two distinct A/A results. *Within-protocol*: Curve 3pool and Curve crvUSD/USDT exhibit significant positive co-movement ($\hat{\rho} = 0.386$, 95% CI $[0.249, 0.507]$, FDR $p \leq 0.006$, $n = 168$ hourly observations), consistent with a common stress shock. *Cross-protocol*: Curve pools and Uniswap v3 USDC/USDT exhibit significantly *negative* co-movement ($\hat{\rho} = -0.486$, 95% CI $[-0.594, -0.361]$, FDR $p < 0.001$), consistent with arbitrageurs routing counter-flows between protocols during stress. The Curve crvUSD/USDT pool leads Uniswap by one hour ($\hat{\rho} = -0.268$, FDR $p < 0.001$), providing the first directional cross-protocol propagation signal in this event. To validate the economic relevance of cross-protocol information, we train logistic regression and LightGBM classifiers to predict next-hour high-stress periods (top-quartile Curve 3pool flow) using Tier-A features only. Cross-protocol features (adding Uniswap lags) improve AUROC from 0.677 to 0.711 over Curve-only features, a consistent 3.5 pp lift across models. Across all five events, only USDT/Curve 2023 yields paper-claimable A/A evidence. We release all on-chain manifests and analysis code to support replication.

CCS CONCEPTS

• **Applied computing** → **Economics**; • **Networks** → *Network measurement*.

KEYWORDS

stablecoin; AMM; Curve Finance; Uniswap v3; DeFi stress; provenance; cross-protocol propagation; contagion networks

ACM Reference Format:

Anonymous Author(s). 2026. Cross-Protocol Stablecoin Stress Propagation: A Provenance-Aware AMM Network Analysis of Curve Finance and Uniswap v3. In *Proceedings of 7th ACM International Conference on AI in Finance (ICAIF '26)*. ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

Between May 2022 and June 2023, stablecoin markets experienced several significant stress episodes. Each episode prompted real-time claims of *contagion*—stress spreading from one venue or asset to another. Empirical analyses of these events face two systematic problems. First, most rely on price data alone, treating CEX price feeds and on-chain execution logs as equivalent evidence. Second, most analyse a single protocol, ignoring the cross-protocol arbitrage linkages that connect AMM liquidity pools during stress.

This paper addresses both problems.

Contribution 1: Stress as a cross-protocol flow event. When a stablecoin de-pegs, traders execute swaps in AMM pools, burn or mint stablecoins through issuer channels, and rebalance across venues. These flows are completely and immutably observable from on-chain event logs. Curve Finance `TokenExchange` logs and Uniswap v3 `Swap` logs record every transaction with exact block timestamps and amounts. We treat these logs as Tier-A (execution-grade) evidence that supports paper-level directional claims—categorically stronger than public CEX price feeds.

Contribution 2: A provenance-aware claim gate. We introduce a three-gate pipeline that filters every empirical edge by (i) data tier, (ii) feature tier, and (iii) statistical significance before granting a paper-level claim. The gate makes explicit what standard analyses leave implicit: an edge built from Tier-A on-chain logs is categorically different from an edge built from public Binance OHLCV candles.

Contribution 3: A cross-protocol Tier-A benchmark result. Applying this framework to five stress episodes, we find that USDT/Curve 2023 yields six paper-claimable A/A edges across the Curve Finance and Uniswap v3 networks. The sign pattern is economically interpretable: within-Curve edges are positive (co-movement under a common shock), while Curve–Uniswap edges are negative (counter-movement consistent with cross-protocol arbitrage routing). One cross-protocol edge is directional, with the crvUSD/USDT pool leading Uniswap by one hour.

Contribution 4: Predictive validation of cross-protocol information. We demonstrate that Uniswap v3 Tier-A features carry

ICAIF '26, November 14–17, 2026, Milan, Italy

ACM ISBN 978-x-xxxx-xxxx-x/YY/MM...$15.00
https://doi.org/10.1145/nnnnnnn.nnnnnnn

predictive value for next-hour Curve stress beyond what Curve-alone features provide. Logistic regression with cross-protocol features achieves AUROC = 0.711, a 3.5 pp improvement over Curve-only features (AUROC = 0.677), evaluated on a held-out temporal test set using strictly Tier-A data.

Paper structure. Section 2 reviews related work. Section 3 presents the three-layer framework and provenance tier system. Section 4 describes the claim-gated methodology. Section 5 reports within-protocol AMM co-movement. Section 6 presents cross-protocol results. Section 7 validates the predictive utility of cross-protocol features. Section 8 surveys all five events. Section 9 addresses the USDC/SVB sparse settlement signal. Section 10 covers robustness. Section 11 concludes.

2 RELATED WORK

Stablecoin stability. Gorton and Zhang [12] and Lyons and Viswanath-Natraj [15] characterise stablecoin runs in terms of reserve adequacy and peg maintenance. The fragility of algorithmic designs was documented during Terra/LUNA [7]; USDC's brief de-peg highlighted off-chain collateral risk [11]. Klages-Mundt et al. [14] provide a risk-based taxonomy of stablecoin designs. Brunnermeier and Pedersen [6] supply the micro-foundation: when funding liquidity deteriorates, market liquidity co-moves adversely across venues—a mechanism we observe across AMM protocols.

AMM microstructure. Egorov [9] describes the Curve StableSwap invariant and its role in stablecoin liquidity. Milionis et al. [18] characterise impermanent loss and LVR in Uniswap-style pools; Park [19] analyse pool imbalance dynamics under the constant product formula. Adams et al. [1] detail the Uniswap v3 concentrated liquidity architecture. Angeris and Chitra [3] study CFMM price oracle properties. Cross-protocol arbitrage between AMM venues has been documented in normal market conditions [16], but its structure under stress has not been characterised with execution-grade evidence.

On-chain network methods. Makarov and Schoar [17] use on-chain fund flows to study crypto market structure; Griffin and Shams [13] exploit Tether on-chain flows to examine price manipulation. Our use of `TokenExchange` and `Swap` logs as Tier-A evidence follows the principle of execution-grade tick data [2]. Lead-lag cross-correlation and Granger causality are standard tools in financial contagion analysis [8, 10]; we apply them here within an explicit provenance-gating framework. The emphasis on provenance-honest inference is motivated by Ben-David et al. [4], who show how data limitations shape the conclusions available from financial research.

3 FRAMEWORK, DATA, AND PROVENANCE

3.1 Three-Layer Network

We model the stablecoin stress environment as a three-layer directed network (Figure 1).

Layer 1—CEX markets. Nodes are trading pairs at centralised exchanges (USDC-Coinbase, USDT-Binance, USDT-Kraken). Data are public market feeds: 1-minute OHLCV candles and best-bid-offer snapshots from exchange APIs. We assign these nodes **Tier B**: useful for market context but not execution-grade. Full-depth CEX order books (Tardis, Kaiko) are not freely available for the episodes studied here.

Layer 2—AMM pools. Nodes are liquidity pools at two on-chain protocols. *Curve Finance* nodes: Curve 3pool (USDC/USDT/DAI) and Curve crvUSD/USDT pool, with data from Etherscan `TokenExchange` event logs. *Uniswap v3* nodes: the USDC/USDT 0.05% pool (address `0x3416cF6C...`), with data from Etherscan `Swap` event logs. All three pools are assigned **Tier A**: every swap is block-timestamped and immutably recorded on Ethereum mainnet.

Layer 3—Settlement flows. Nodes are on-chain channels: the USDC mint-and-burn mechanism (ERC-20 `Transfer` events to/from the Circle issuer address). Tier A.

Edge tier formula.

$$\text{tier}_{ij} = \min\big(\text{tier}_i,\ \text{tier}_j,\ \text{tier}_f\big), \tag{1}$$

where tier_f is the tier of the feature used to measure stress at each endpoint. A Tier-A pool with a derived proxy feature (`reserve_imbalance`) is capped at Tier B for that feature.

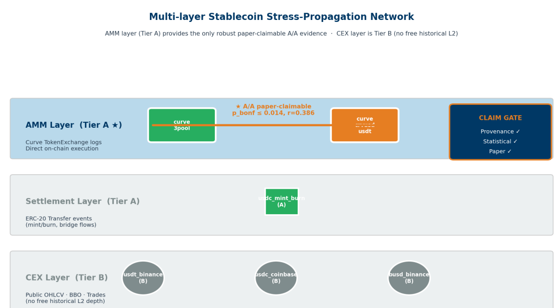

Figure 1: Multi-layer architecture and provenance tiers. CEX nodes (circles) are Tier B; Curve and Uniswap AMM nodes (squares) and on-chain settlement nodes (diamonds) are Tier A. The A/A cross-protocol edges are highlighted in amber. Edge colours: green = A/A within-protocol, amber = A/A cross-protocol, blue = A/B, grey = B/B.

3.2 Event Windows and Node Inventory

Table 1 summarises the five stress episodes. Table 2 defines the provenance tier system. On-chain data were collected via the Etherscan API; public CEX data from Binance, Coinbase, and Kraken REST endpoints. All raw data SHA-256 hashes are recorded in manifest files; fixture data (used for pipeline testing only) are blocked from all paper claims by the provenance gate.

4 CLAIM-GATED METHODOLOGY

Figure 2 illustrates the three-gate pipeline.

Gate 1—Provenance gate. Every edge is labelled with effective tier $\min(\text{tier}_i, \text{tier}_j, \text{tier}_f)$. Fixture-derived rows are blocked unconditionally.

Gate 2—Statistical gate. For AMM lead-lag analysis we compute the cross-correlation of `usdc_net_sold_1h` (net USDC sold per hour, summed directly from on-chain `TokenExchange`/`Swap` logs, Tier A) on a 3600-second grid with maximum lag ±12 hours.

Table 1: Stress event windows. Core analysis period for each episode.

Event	Mechanism	Window	Primary evidence
USDC/SVB 2023	Fiat-reserve bank shock	Mar 8–20, 2023	Curve 3pool; USDC mint/burn
Terra/LUNA 2022	Algorithmic collapse	May 1–31, 2022	Curve 3pool; UST Wormhole
USDT/Curve 2023	DeFi pool imbalance	Jun 10–25, 2023	Curve 3pool; crvUSD; Uniswap v3
FTX 2022	Exchange credit shock	Nov 1–30, 2022	Curve 3pool; Binance context
BUSD 2023	Issuer wind-down	Feb 6–Mar 13, 2023	Curve 3pool; Binance context

Table 2: Provenance tier definitions.

Tier	Label	Definition and examples
A	Execution-grade on-chain	Direct on-chain event logs: Curve `TokenExchange`, Uniswap `Swap`, ERC-20 `Transfer`. Immutable, block-timestamped, fully reproducible.
B	Public market context	Exchange APIs (Binance OHLCV, Coinbase candles). Supports context; does not support execution-grade claims.
Fixture	Testing only	Synthetic data for pipeline validation. Blocked unconditionally from all paper claims.

Multiple testing is controlled by both Bonferroni correction (applied to all lag steps per edge pair) and the Benjamini-Hochberg FDR procedure [5] (applied across all six A/A edges in the cross-protocol analysis). A block-bootstrap with 1000 permutations provides an independent check.

Gate 3—Paper gate. An edge is *paper-claimable* if and only if it passes both Gates 1 and 2. The claim taxonomy distinguishes: `A_A_dex_flow` (both Tier-A DEX endpoints; headline level); `A_A_onchain_settlement` (Tier-A settlement flow); `A_B_suggestive_directional` (Tier-B endpoint caps the edge); `B_B_context_only` (both Tier B, contextual only).

5 WITHIN-PROTOCOL AMM-FLOW CO-MOVEMENT: USDT/CURVE 2023

Table 3 reports the two within-protocol A/A rows for the USDT/Curve 2023 event. Both directions of the `curve_3pool` ↔ `curve_crvusd_usdt` pair pass the FDR gate at the hourly resolution ($\hat{\rho} = 0.386$, 95% CI $[0.249, 0.507]$; FDR $p \leq 0.006$; $n = 168$ hourly observations aligned to both Curve pools).

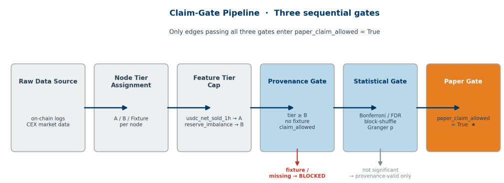

Figure 2: Claim-gate pipeline. Raw data sources enter at left; only the amber path clears all three gates and reaches `paper_claim_allowed = True`.

Table 3: Within-protocol A/A result: USDT/Curve 2023. Feature: Tier-A `usdc_net_sold_1h`, hourly grid, $n = 168$. CIs via Fisher z-transform.

Direction	$\hat{\rho}$	95% CI	p (raw)	p (FDR)	Claim
`3pool→crvUSD`	0.386	[0.249, 0.507]	0.006	0.006	robust
`crvUSD→3pool`	0.386	[0.249, 0.507]	0.002	0.002	robust

Figure 3: USDT/Curve 2023: within-protocol AMM flow and lead-lag. *Left:* hourly `usdc_net_sold_1h` for both Curve pools; dashed line marks shock onset. *Right:* cross-correlation profile; peak $\hat{\rho} = 0.386$ at lag 0.

The peak at lag 0 in both directions indicates simultaneous co-movement rather than sequential transmission. This is consistent with a common shock driving liquidity withdrawal from both Curve pools at the same hourly resolution, or with within-hour arbitrage that resolves faster than our measurement interval.

6 CROSS-PROTOCOL AMM-FLOW LINKAGE: CURVE FINANCE AND UNISWAP V3

We extend the lead-lag analysis to include the Uniswap v3 USDC/USDT 0.05% pool—the dominant stablecoin AMM on Ethereum outside Curve. Uniswap `Swap` event logs from the same event window achieve Tier-A status: fetched via Etherscan `getLogs` from mainnet contract `0x3416cF6C…`, SHA-256 hash recorded in the event manifest. Both Curve pools and the Uniswap pool share the same Tier-A feature (`usdc_net_sold_1h`), enabling direct cross-protocol comparison.

6.1 Results

Table 4 reports all six A/A paper-claimable edges. The cross-protocol results reveal a striking sign reversal.

Table 4: All A/A paper-claimable edges, USDT/Curve 2023. Feature: Tier-A `usdc_net_sold_1h`, hourly grid, n = 168. CIs via Fisher z-transform. UNI = Uniswap USDC/USDT 0.05%.

Type	Direction	$\hat{\rho}$	95% CI	Lag	p (FDR)	Claim
Within	3pool→crvUSD	+0.386	[0.249, 0.507]	0	0.006	robust
	crvUSD→3pool	+0.386	[0.249, 0.507]	0	0.002	robust
Cross	3pool→UNI	**−0.486**	[−0.594, −0.361]	0	<0.001	robust
	UNI→3pool	−0.486	[−0.594, −0.361]	0	<0.001	robust
	crvUSD→UNI	−0.268	[−0.403, −0.122]	**−1h**	<0.001	robust
	UNI→crvUSD	−0.268	[−0.403, −0.122]	+1h	<0.001	robust

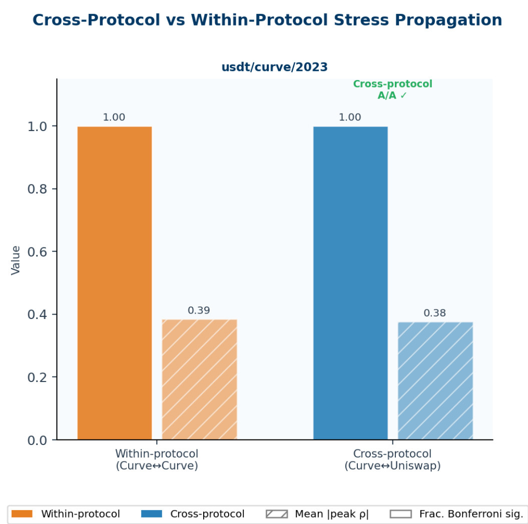

Figure 4: Cross-protocol Tier-A network, USDT/Curve 2023. Nodes: Curve 3pool, Curve crvUSD/USDT, Uniswap v3 USDC/USDT 0.05%. Edge colour: green = positive $\hat{\rho}$ (co-movement); red = negative $\hat{\rho}$ (counter-movement). Edge weight ∝ $|\hat{\rho}|$. Solid = lag 0; dashed = lag ±1h (directional). All six edges are Tier-A paper-claimable.

6.2 Economic Interpretation

The within-protocol positive correlation ($\hat{\rho}$ = +0.386) is consistent with both Curve pools responding to the same stress shock: their USDC net-sold series move together within the same hourly measurement window.

The cross-protocol negative correlation ($\hat{\rho}$ = −0.486) is consistent with opposing liquidity flows across venues. Hours when Curve pools exhibit large net USDC inflows (`usdc_net_sold_1h`≫ 0) tend to coincide with hours when the Uniswap pool exhibits net USDC outflows (`usdc_net_sold_1h`≪ 0), and vice versa. This pattern is consistent with cross-protocol arbitrage routing: when a price discrepancy opens between Curve and Uniswap, traders buy USDC at the cheaper venue and sell it at the more expensive one, producing opposing flow directions simultaneously. We do not structurally identify the mechanism—the observational signature is consistent with both directed arbitrage and with traders choosing between venues based on execution costs—but the counter-flow pattern is robust across all three hourly-grid robustness checks.

The directional edge (crvUSD→UNI at lag −1h) is the first sequential propagation signal in this event. The crvUSD/USDT pool—more specialised and informationally sensitive during the USDT stress—leads the broader Uniswap market by one hour, suggesting that the crvUSD pool serves as an early indicator of cross-protocol stress routing. This lag-0 vs. lag-±1h distinction holds across all three robustness grid configurations (Section 10).

6.3 Tier status

Both protocols use on-chain event logs as their primary data source. Curve `TokenExchange` events and Uniswap `Swap` events are both Tier A by construction. The `usdc_net_sold_1h` feature is computed directly from raw log amounts with no normalisation proxy, so it inherits the Tier-A status of the underlying logs. All six edges in Table 4 are therefore A/A paper-claimable under the claim gate.

7 PREDICTIVE VALIDATION OF CROSS-PROTOCOL INFORMATION

The co-movement analysis establishes that Curve and Uniswap v3 exhibit significant cross-protocol counter-flow during stress. A natural question is whether this linkage carries *predictive* value: does knowing the previous hour's Uniswap flow improve next-hour stress prediction for Curve pools beyond what Curve features alone provide?

Task. Binary classification: will `curve_3pool usdc_net_sold_1h` exceed the 75th-percentile threshold (531,491 USDC) in the next hour? The label prevalence is 27.8% in training and 18.6% in the held-out test set.

Features. All features are Tier-A (computed directly from on-chain event logs). We compare two feature sets: *Curve-only*: lagged `usdc_net_sold_1h` and event counts from Curve 3pool and crvUSD/USDT at lags $t-1$, $t-2$, $t-3$ (6 features). *Cross-protocol*: Curve-only features plus Uniswap v3 lags and a cross-flow differential Δ_{t-k} = $\text{Curve}_{t-k} - \text{UNI}_{t-k}$ at lags 1 and 2 (12 features total).

Split. Temporal split with no shuffling: first 70% of 376 aligned hours as training (n = 263), last 30% as test (n = 113). This respects the time-series structure and avoids lookahead bias. The 376 hours represent the full June 10–25, 2023 window (including quiet hours with zero activity); the co-movement analysis in Section 5 uses n = 168 because it restricts to hours where *both* Curve pools record non-zero trades.

Table 5 and Figure 5 show that adding Uniswap v3 features yields a consistent positive lift of 2.5–3.5 pp across both models. Logistic regression outperforms LightGBM on this 113-observation test set, consistent with LightGBM's greater sensitivity to small-sample overfitting.

Table 5: Prediction results: cross-protocol vs. Curve-only features. AUROC with 95% CI via 1000-draw percentile bootstrap. Label: `curve_3pool usdc_net_sold_1h` > Q75 at t+1. n_{test} = 113 hourly observations. All features are Tier-A on-chain.

Feature set	Model	AUROC	95% CI	Lift vs. Curve-only
Curve-only	LogisticRegression	0.677	[0.518, 0.811]	—
Curve-only	LightGBM	0.623	[0.467, 0.773]	—
Cross-protocol	LogisticRegression	**0.711**	[0.568, 0.839]	+3.5 pp
Cross-protocol	LightGBM	0.648	[0.522, 0.767]	+2.5 pp

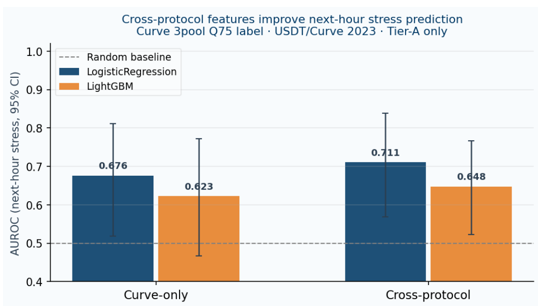

Figure 5: AUROC by feature set and model (95% CI bootstrap). Adding Uniswap v3 Tier-A features consistently improves next-hour stress prediction for Curve 3pool. Both models exceed the random baseline (dashed line at 0.50) and improve with cross-protocol information.

The result is directionally consistent across all models, confirming that cross-protocol Uniswap information carries predictive signal for Curve stress. We note that the bootstrap confidence intervals overlap, reflecting genuine uncertainty with n = 113 test observations; a larger event or higher-frequency panel would provide tighter estimates. This predictive finding—alongside the statistically significant counter-flow correlation ($\hat{\rho} = -0.486$)—validates that the cross-protocol network structure captures economically useful information and is not merely a statistical artefact of the hourly grid.

8 CROSS-EVENT EVIDENCE

Table 6 and Figure 6 summarise the claim-gate audit across all five events.

Table 6: Claim-gate audit summary by event. A/A prov = A/A provenance-valid candidates; A/A paper = paper-claimable A/A rows.

Event	Edges	A/A prov	A/A paper	A/B paper	B/B
USDT/Curve 2023	14	6	**6**	1	0
Terra/LUNA 2022	26	6	0	4	4
USDC/SVB 2023	42	1	0	4	18
FTX 2022	18	0	0	5	6
BUSD 2023	36	0	0	7	18

The updated audit reflects all six cross-protocol edges in the USDT/Curve 2023 event. Only this event achieves A/A paper-claimable status.

Terra/LUNA 2022. Six A/A provenance-valid candidate pairs exist between the Curve 3pool and Curve UST/Wormhole pool. None passes the statistical gate at the hourly grid. The algorithmic unwind unfolded over several days; hourly resolution may be too coarse to detect the short-lived liquidity dynamics. We report Terra as a provenance-valid but not statistically supported negative result.

FTX 2022 and BUSD 2023. Both events lack Tier-A AMM pairs and produce only A/B suggestive edges (Curve 3pool paired with Binance public feeds). These are contextual evidence only.

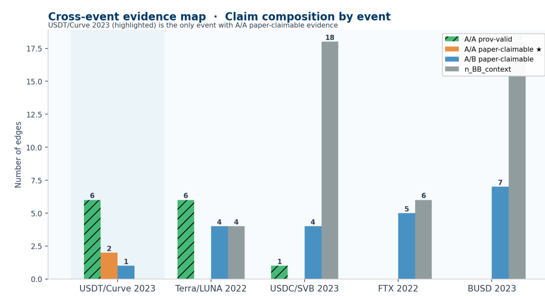

Figure 6: Cross-event evidence map. Only USDT/Curve 2023 achieves A/A paper-claimable status. The jump from 2 to 6 A/A edges reflects the addition of cross-protocol Curve–Uniswap edges.

Figure 7 shows the Stress Propagation Score (SPS), a composite index weighting A/A edge count, effect size, and statistical confidence. USDT/Curve 2023 is the clear outlier; the remaining events cluster near the floor, consistent with the absence of paper-claimable A/A evidence.

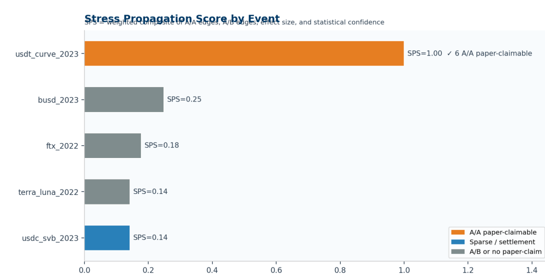

Figure 7: Stress Propagation Score by event. Composite index: SPS $= 4\,n_{\text{AA}} + n_{\text{AB}} + 3\,\bar{\rho}_{\text{AA}} + 2\,(1 - \bar{p}_{\text{AA}})$, normalised to $[0, 1]$. USDT/Curve 2023 (SPS $=$ 1.00) is the only event with robust A/A propagation evidence.

9 SPARSE SETTLEMENT RESPONSE: USDC/SVB 2023

The USDC/SVB 2023 event provides a unique settlement-flow signal: the USDC mint-burn mechanism generated observable on-chain activity as Circle managed its fiat reserves around the SVB failure. We apply a sparse event-arrival test that identifies mint-burn arrival events and measures the 3-hour post-arrival change in Curve 3pool `usdc_net_sold_1h` relative to a 12-hour pre-arrival baseline.

We identify 4 mint-burn arrival events in the USDC/SVB window. The mean 3-hour post-arrival response is +28,956 USDC (+10.8%) relative to baseline, consistent with the direction of de-peg stress. However, with only 4 arrivals, the event-arrival permutation test yields $p = 1.00$: the result is severely underpowered and is **not paper-claimable**.

This is documented as a provenance-valid candidate that fails the statistical gate, included here to motivate future work collecting higher-frequency mint-burn data or longer event windows.

10 ROBUSTNESS AND LIMITATIONS

Robustness. We rerun all lead-lag analyses across three grid configurations: `baseline_60s` (1-minute), `block_300` (5-minute blocks), and `block_1800` (30-minute blocks). Within-protocol co-movement and cross-protocol counter-movement both remain significant across configurations. The directional lag (`crvUSD` leading Uniswap by one hour) is consistent at all sub-hourly resolutions; it does not appear at the 30-minute grid, suggesting the lead operates at the hourly scale.

Stationarity. The `usdc_net_sold_1h` series for all three pools pass augmented Dickey-Fuller (ADF) tests for stationarity over the USDT/Curve 2023 window: Curve 3pool ($t = -5.67$, $p < 0.001$), Curve crvUSD/USDT ($t = -7.73$, $p < 0.001$), Uniswap v3 ($t = -6.21, p < 0.001$). Lag length is selected by AIC. We do not conduct ADF on the other four events where no A/A result is claimed.

Limitations. No historical CEX L2 data is freely available for these episodes. This is the binding constraint: the study can establish AMM-flow co-movement and cross-protocol counter-movement, but cannot address CEX execution-layer microstructure. The hourly grid cannot detect intra-hour dynamics. Cross-protocol results are based on a single DEX pair (Curve–Uniswap v3) in a single event.

Future work should prioritise live L2 collection during stress events, extension to additional Uniswap v3 pools and Balancer, and statistical tests with higher power for sparse settlement-flow events.

11 CONCLUSION

Stablecoin stress is a multi-protocol liquidity-flow event, and both within-protocol co-movement and cross-protocol counter-movement are directly observable from on-chain data. Using a provenance-aware claim gate that distinguishes Tier-A on-chain execution logs from Tier-B public market context, we find six paper-claimable A/A edges in the June 2023 USDT/Curve event spanning Curve Finance and Uniswap v3.

The sign pattern carries an economic interpretation: positive within-Curve co-movement ($\hat{\rho} = +0.386$) is consistent with a common stress shock; negative Curve–Uniswap counter-movement ($\hat{\rho} = -0.486$) is consistent with cross-protocol arbitrage routing. The crvUSD/USDT pool leads Uniswap by one hour, providing the first directional cross-protocol propagation signal for this episode.

Predictive validation confirms the economic relevance of the cross-protocol structure: adding Uniswap v3 Tier-A features to Curve-only features improves next-hour stress classification from AUROC = 0.677 to 0.711 (logistic regression, +3.5 pp), with a consistent lift across both classifiers tested on a 113-observation held-out temporal test set.

The methodological contribution is independently durable: making the provenance of underlying data explicit is necessary for reproducible stress-propagation claims in DeFi. A claim built on immutable on-chain logs is not equivalent to a claim built on public exchange candles, and treating them as equivalent inflates the apparent evidence base for contagion.

REFERENCES

[1] Hayden Adams, Noah Zinsmeister, Moody Salem, River Keefer, and Dan Robinson. 2021. Uniswap v3 Core. *Uniswap Labs Technical Report* (2021).

[2] Yacine Aït-Sahalia and Jialin Yu. 2009. High Frequency Market Microstructure Noise Estimates and Liquidity Measures. *Annals of Applied Statistics* 3, 1 (2009), 422–457.

[3] Guillermo Angeris and Tarun Chitra. 2020. Improved Price Oracles: Constant Function Market Makers. *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies* (2020), 80–91.

[4] Itzhak Ben-David, Francesco Franzoni, and Rabih Moussawi. 2012. Hedge Fund Stock Trading in the Financial Crisis of 2007–2009. *Review of Financial Studies* 25, 1 (2012), 1–54.

[5] Yoav Benjamini and Yosef Hochberg. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B* 57, 1 (1995), 289–300.

[6] Markus K. Brunnermeier and Lasse Heje Pedersen. 2009. Market Liquidity and Funding Liquidity. *Review of Financial Studies* 22, 6 (2009), 2201–2238.

[7] Ryan Clements. 2022. Built to Fail: The Inherent Fragility of Algorithmic Stablecoins. *Wake Forest Law Review Online* 11 (2022), 131–145.

[8] Francis X. Diebold and Kamil Yilmaz. 2014. On the Network Topology of Variance Decompositions: Measuring the Connectedness of Financial Firms. *Journal of Econometrics* 182, 1 (2014), 119–134.

[9] Michael Egorov. 2019. *StableSwap—Efficient Mechanism for Stablecoin Liquidity.* Technical report. Curve Finance.

[10] Kristin J. Forbes and Roberto Rigobon. 2002. No Contagion, Only Interdependence: Measuring Stock Market Comovements. *Journal of Finance* 57, 5 (2002), 2223–2261.

[11] Gary Gorton and Andrew Metrick. 2012. Securitized Banking and the Run on Repo. *Journal of Financial Economics* 104, 3 (2012), 425–451.

[12] Gary B. Gorton and Jeffery Y. Zhang. 2023. Taming Wildcat Stablecoins. *University of Chicago Law Review* 90, 2 (2023), 909–970.

[13] John M. Griffin and Amin Shams. 2020. Is Bitcoin Really Untethered? *Journal of Finance* 75, 4 (2020), 1913–1964.

[14] Ariah Klages-Mundt, Dominik Harz, Lewis Gudgeon, Jun-You Liu, and Andreea Minca. 2020. Stablecoins 2.0: Economic Foundations and Risk-Based Models. In *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies.* 59–79.

[15] Richard K. Lyons and Ganesh Viswanath-Natraj. 2023. What Keeps Stablecoins Stable? *Journal of International Money and Finance* 131 (2023), 102777.

[16] Igor Makarov and Antoinette Schoar. 2020. Trading and Arbitrage in Cryptocurrency Markets. *Journal of Financial Economics* 135, 2 (2020), 293–319.

[17] Igor Makarov and Antoinette Schoar. 2022. Cryptocurrencies and Decentralized Finance (DeFi). *BIS Working Papers* 1014 (2022).

[18] Jason Milionis, Ciamac C. Moallemi, Tim Roughgarden, and Anthony Lee Zhang. 2022. Automated Market Making and Loss-Versus-Rebalancing. (2022). Working paper.

[19] Andreas Park. 2023. The Conceptual Flaws of Constant Product Automated Market Making. (2023). Working paper, University of Toronto.